

PHYS-111 MECHANICS AND THERMAL PROPERTIES OF MATTER

ODL END OF SEMESTER

[100 MARKS]

QUESTION 1

- The position of a particle in space is described by (3.00, 4.00) m. find the polar coordinate of this particle.
- The speed, v , of an object is given by the equation $v = At^3 - Bt$, where t refers to time. What is the dimension of A and B ?
- States **two** major application of dimensional analysis.
- King Kong carries Fay Wray up the 321m tall Empire State Building. At the top of the skyscraper, Fay Wray's shoe falls from her foot. How fast will the shoe be moving when it hits the ground?
- If St. Louis Cardinals homerun king, Mark McGwire, hit a baseball due west with a speed of 50.0 m/s, and the ball encountered a wind that blew it north at 5.00 m/s, what was the resultant velocity of the baseball?
- Define the term projectile?
- In her physics lab, Melanie rolls a 10-g marble down a ramp and off the table with a horizontal velocity of 1.2 m/s. The marble falls in a cup placed 0.51 m from the table's edge. How high is the table?

QUESTION 2

- A 30.0-g arrow is shot by William Tell through an 8.00-cm-thick apple sitting on top of his son's head. If the arrow enters the apple at 30.0 m/s and emerges at 25.0 m/s in the same direction, with what force has the apple resisted the arrow?

- b. Rose is sledding down an ice-covered hill inclined at an angle of 15° with the horizontal. If Rose and the sled have a combined mass of 54.0 kg, what is the force pulling them down the hill?
- c. At Sea World, a 900.-kg polar bear slides down a wet slide inclined at an angle of 25.0° to the horizontal. The coefficient of friction between the bear and the slide is 0.0500. What frictional force impedes the bear's motion down the slide?
- d. After returning home from the beach, Samantha hangs her wet 0.20-kg bathing suit in the center of the 6.0-m-long clothesline to dry. This causes the clothesline to sag 4.0 cm. What is the tension in the clothesline?
- e. A tennis ball traveling at 10.0 m/s is returned by Venus Williams. It leaves her racket with a speed of 36.0 m/s in the opposite direction from which it came. a) What is the change in momentum of the tennis ball? b) If the 0.060-kg ball is in contact with the racket for 0.020 s, with what average force has Venus hit the ball?
- f. Charlotte, a 65.0-kg skin diver, shoots a 2.0-kg spear with a speed of 15 m/s at a fish who darts quickly away without getting hit. How fast does Charlotte move backwards when the spear is shot?

QUESTION 3

- a. State the law of conservation of energy
- b. After finishing her physics homework, Sherita pulls her 50.0-kg body out of the living room chair and climbs up the 5.0-m-high flight of stairs to her bedroom. How much work does Sherita do in ascending the stairs?
- c. In a wild shot, Bo flings a pool ball of mass m off a 0.68-m-high pool table, and the ball hits the floor with a speed of 6.0 m/s. How fast was the ball moving when it left the pool table? (Use the law of conservation of energy.)
- d. Curtis' favorite disco record has a scratch 12 cm from the center that makes the record skip 45 times each minute. What is the linear speed of the scratch as it turns?
- e. State the law of universal gravitation

QUESTION 4

- a. The layer of the sun that we see is called the photosphere. It has a temperature of 5600 K. What is the sun's temperature a) in degrees Celsius? b) in degrees Fahrenheit?
- b. Define specific heat.
- c. An igloo is made of 224 blocks of ice at 0°C , each with a mass of 12.0 kg. How much heat must be gained by the ice to melt the entire igloo?
- d. Gwyn's bowl is filled with 0.175 kg of 60.0°C soup (mostly water) that she stirs with a 20.0°C silver spoon of mass 0.0400 kg. The spoon slips out of her hand and slides into the soup. What equilibrium temperature will be reached if the spoon is allowed to remain in the soup and no heat is lost to the outside air? (cspoon = $240. \text{ J/kg}^{\circ}\text{C}$) Assume that the temperature of the bowl does not change.